Shweta Sarap

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SUMMARY

Highly motivated engineer with a strong foundation in software development and hardware verification, seeking to apply expertise in designing and implementing automation solutions for complex hardware designs. Passionate about developing efficient methodologies and tools to enhance design processes and driven to solve challenging technical problems.

PROFESSIONAL EXPERIENCE

Teachers Insurance and Annuity Association (TIAA): QA Associate Jan 2022 - May 2024

- Led debugging and defect analysis, collaborating with developers to isolate issues, perform root cause analysis, and ensure design and functional coverage met verification standards.
- Tracked defects using JIRA and ALM, improving resolution time by 50% through systematic debugging, failure analysis, and validation of system behavior against specifications.
- Conducted functional, end-to-end, and boundary condition testing, ensuring specification compliance, correctness across different operating conditions, and robustness under edge cases.
- Developed and integrated automated verification scripts, optimizing regression testing and enhancing validation efficiency through automation frameworks.
- Collaborated with cross-functional teams to streamline verification workflows, ensuring smooth integration of test environments and alignment with validation methodologies.

Atos Syntel: Associate Consultant Nov 2018 - Jan 2022

- Developed and executed verification plans, ensuring functional correctness and meeting design specifications.
- Debugged and analyzed verification results, identifying potential design flaws & improving hardware validation efficiency.
- Applied formal verification techniques, including Model Checking and SystemVerilog Assertions (SVAs), to verify design functionality and correctness.
- Developed modeling code to support formal verification and validate design functionality.
- Optimized regression methodologies, automating test execution, failure tracking, and reporting to improve verification coverage.
- Assisted in the development and maintenance of verification scripts.
- Applied formal techniques such as Abstractions and Design Reductions.
- Optimized Formal Verification (FV) infrastructure, standardizing processes, applying abstractions, and design reductions to improve efficiency.
- Created detailed documentation of verification processes, test results, and known issues for future reference.

TECHNICAL SKILLS

- **Tools:** JasperGold ,VC Formal, Superlint, EDA tools, JIRA, GIT, Jenkins
- **Programming Languages:** Python Scripting, Java, C/C++, System Verilog, System Verilog Assertions, TCL, Shell Scripting
- **Methodologies:** ASIC design & RTL Verification**,** Universal Verification Methodology (UVM)

- **Testing Techniques :** Functional, Regression Testing, Boundary Value Analysis, Defect Management, Bug Tracking, Root Cause Analysis , Bug Fix Verification
- **Formal Verification:** JasperGold FPV App, Coverage App, sequential Equivalence Checking (SEC) App, Counter Abstractions, Preloading (IVAs), Non-deterministic Variables, Jasper Data Scoreboards, Engine Optimizations, Connectivity Apps

EDUCATION

M.S. Electrical Engineering

Arizona State University, Tempe, AZ, USA — 4.00 GPA

Coursework: Semiconductor Device Theory, Software Process & Quality Management, Verification & Testing, Semiconductor Process/Device Simulation, Fundamentals of Semiconductor Packaging

M. Tech Computer Engineering

University of Pune, Pune, MH, India — 3.20 GPA

B.E Computer Engineering

University of Pune, Pune, MH, India — 3.40 GPA